📘 MODULE 19: CONFIGURATION PRECEDENCE

19.1 Concept

The same setting can appear in many files. Splunk must decide **which value wins**. It merges all layers and **higher-priority layers override lower ones, attribute by attribute**.

19.2 The layers (simplified)

Global context (index-time, inputs, server settings):

```
1. etc/system/local          ← wins
2. etc/apps/<app>/local
3. etc/apps/<app>/default
4. etc/system/default        ← lowest
```

Search-time context (app/user, e.g. extractions, lookups, saved searches):

```
1. etc/users/<user>/<app>/local   ← wins
2. etc/apps/<current app>/local
3. etc/apps/<current app>/default
4. other apps' settings (only if shared globally)
5. etc/system/local
6. etc/system/default             ← lowest
```

Within the same app: `local` beats `default`.
Between different apps (global context): app directory names are compared in ASCII order, so an app named `A_app` beats `B_app`. This matters when two add-ons configure the same thing. The exact rules are subtle, so use `btool` instead of guessing.

19.3 WHY

- `default` holds the vendor's settings, and updates **overwrite** it.
- `local` holds yours, and updates **never touch it**.
- So `local` must win, and that is why you never edit `default`.

19.4 Merging example

`system/default/props.conf` has:

ini

```ini
[linux_secure]
SHOULD_LINEMERGE = false
TRUNCATE = 10000
```

`apps/search/local/props.conf` has:

ini

```
[linux_secure]
TRUNCATE = 50000
```

Effective result: `SHOULD_LINEMERGE = false` (from default) and `TRUNCATE = 50000` (local wins). Only the **conflicting setting** is overridden. The rest is merged.

19.5 Troubleshooting with btool

bash

```
sudo /opt/splunk/bin/splunk btool props list linux_secure --debug
```

- `props list linux_secure` shows the merged `[linux_secure]` stanza.
- `--debug` shows the **file path** for each setting, so you can see **which layer won**.
- Look for a setting coming from an unexpected app.

Practical troubleshooting scenarios

Scenario	How to troubleshoot
"My change isn't working"	`btool ... --debug`: is **your** value the effective one? If another file wins, fix the layer or the app order.
Change is effective in btool but not in behavior	It needs a **restart** (index-time) or you edited the wrong **tier** (e.g. a props parsing change on the UF).
Two add-ons conflict	`btool` shows which app is winning. Adjust in `local` or change app ownership/priority.
A setting disappears after an upgrade	You edited `default`, so move it to `local`.
Works for me, not others	Search-time **user** layer or private permissions.

19.6 Applying changes

Change	Apply
Index-time (inputs, props parsing, transforms, indexes)	**Restart** the instance doing the work
Many search-time settings (props extractions, lookups, tags)	Refresh: `http://<host>:8000/debug/refresh`, or reload with a restart if unsure
Deployment-managed config on forwarders	Deployment server pushes it, and the forwarder may restart

If unsure, restart in a lab. In production, check Splunk docs for the specific setting.

19.7 Hands-on

1. Run `btool props list linux_secure --debug` and note which files contribute.
2. Create `/opt/splunk/etc/apps/search/local/props.conf` with:

```ini
[linux_secure]
TRUNCATE = 20000
```

Restart, then re-run btool. Confirm your value shows with the `local` path.
3. Remove it again and confirm the value returns to default.

19.8 Interview Q&A

- **What is the order of precedence?** User > app local > app default > system local > system default for search-time context. In global context, system/local > app/local > app/default > system/default.
- **Why use `local`?** It survives upgrades and has higher priority than `default`.
- **Are config files replaced or merged?** Merged per setting.
- **How do you find the effective setting?** `splunk btool <conf> list --debug`.
- **Your change isn't applied, what do you check?** btool for winning layer, correct tier and app, restart or refresh, and typos.

19.9 Revision

local beats default · system/local is top in global context · merged per attribute · `btool --debug` shows the winner · index-time changes need a restart.